Question ID: 96467fea

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

Circle N has a radius of 7 millimeters (mm). Circle M has an area of $64\pi\ \ \mathrm{mm}^2$. What is the total area, in mm^2, of circles N and M?

Answer

A. 113π

B. 92π

C. 78π

D. 15π

Correct Answer: A

Rationale

Choice A is correct. It's given that circle N has a radius of $7\ \mathrm{mm}$. The area of a circle is given by the expression πr^2, where r is the radius of the circle. Substituting 7 for r in the expression yields $\pi(7)^2$, or 49π, for the area, in mm^2, of circle N. It's also given that the area of circle M is 64π mm^2. Adding the two areas yields $49\pi + 64\pi$, or 113π, mm^2. Therefore, the total area, in mm^2, of circles N and M is 113π.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.